



Experimental study on mitigating vortex-induced vibration of a bridge by using passive vortex generators



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ABSTRACT

It has been reported that three-dimensional spanwise-varying control is very efficient in suppressing vortex shedding behind bluff bodies. Therefore, a new passive control method, spanwise-varying passive vortex generator (PVG), was proposed to mitigate the vortex-induced-vibration (VIV) of a bridge. The PVGs can directly generate streamwise vortices, which behave like secondary wake instability modes (3D instability modes). Therefore, the PVGs can cause 3D instability of spanwise vortex shedding and suppress VIV. In this study, isolated PVGs were arranged on the lower surface of the bridge in spanwise direction, and their control effects on mitigating VIV were investigated through wind tunnel tests. The results show that as PVGs are arranged with spanwise distance of $1-3H$ (H is the height of the bridge) and the PVG height is $0.1-0.25H$, both vertical and torsional VIV can be completely suppressed, and the trailing edge is the key region to suppress VIV. Further, the control effects of PVG are barely influenced by the boundary layer of the bridge. The spectrum analysis of the wake velocity shows that the PVG can completely suppress the vortex shedding near a wake, and thereby resulting in the vanishing of VIV.

1. Introduction

As wind passes around the long span bridge deck, the spanwise vortices will shed in the near wake. If the vortex shedding frequency is close to the natural frequency of the bridge deck, it can cause vortex-induced resonance. A vortex-induced vibration (VIV) is one of the major issues in long-span bridge vibration. Although the VIV is a self-limited vibration, it can introduce large displacements and cause discomfort to the drivers (Battista and Pfeil, 2000; Frandsen, 2001; Fujino and Yoshitaka, 2002; Larsen et al., 2000). In addition, vortex-induced resonance always occurs at low wind speeds and can lead to long-term fatigue damage.

Many types of control methods for suppressing VIV have been proposed. However, passive flow control methods are usually selected because of their reliability and simplicity. Fujino and Siringoringo (2013) provided a review of these control methods in bridge engineering. Conventional passive flow control attachments, which are always attached to the leading and trailing edges, include guide vanes, fairing, flaps, wind spoiler and deflector (Fujino and Siringoringo, 2013). Among them, guide vanes are more popularly used in suppressing VIV since they

provide better control effect. Larsen et al. (2000) installed guide vanes on the Greatbelt bridge deck, which was then subjected to severe VIV, and observed a reduction in the amplitude of the VIV from the wind tunnel test results. Later, Larsen et al. (2008) investigated the vortex response of a twin box bridge with and without the guide vanes at three different Reynolds numbers and observed that the guide vanes had no obvious control effects on the VIV at low Reynolds number as no sufficient flow rate passed it with high boundary layer. Therefore, it indicated that the control effects of the guide vanes were sensitive to the Reynolds numbers. Further, the aerodynamic countermeasures above always increase the maintenance efforts and construction costs since these attachments need be installed along with the bridge deck. In addition, Larsen and Wall (2012) explored the effect of the shape on the VIV of trapezoidal box girder bridge decks and found that it was possible to derive a virtually vibration-free deck. However, this approach could fail if the shape was restricted because of the other requirements. Additionally, the shape of a vibration-free deck might also not be considered esthetic.

As the regular vortex shedding is the main source of the VIV, suppressing spanwise vortex shedding is an efficient way to mitigate the VIV of a bridge deck. The three-dimensional spanwise-varying control, which

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directly or indirectly changes the spanwise aerodynamic shape of a bluff body, is more efficient than the spanwise homogeneous method (based on 2D framework) on attenuating vortex shedding. Kim and Choi (2005) compared the momentum required for drag reduction of a 3D spanwise-varying control method and a spanwise homogeneous method (base bleeding). The results indicated that the momentum required of the three-dimensional forcing was significantly less than that of a conventional two-dimensional forcing (base bleeding). The underlying mechanism of the three-dimensional spanwise-varying control can be ascribed to the action of the secondary wake instability, which is also called the 3D instability of spanwise vortices (Hwang et al., 2013). There are mainly two types of secondary instabilities existing in a bluff body wake, namely the Modes-A and Modes-B (Williamson, 1996a,b; Wu et al., 1996). Both instability modes act as waves in the spanwise wake vortices, which evolve into pairs of counter-rotating streamwise vortices connecting the Karman vortices. These secondary streamwise vortices cause the spanwise dislocation of the Karman vortices and the energy transfer from the spanwise vortices to the streamwise vortices (Doddipatla et al., 2008). Because of the highly efficient control effects of the 3D spanwise-varying control, its application for mitigating vortex shedding and/or achieving drag reduction has attracted much attention. The numerical simulation results of Darekar and Sherwin (2001) showed that for providing spanwise sinusoidal perturbations at the leading and trailing edges for a square cylinder, the spanwise vortex shedding was highly suppressed at $10 < Re(d) < 150$ (d being the character length of the bluff body) when the perturbation wavelength was $5.6d$. Kim and Choi (2005) investigated the control effects of spanwise-periodic blowing/suction on reducing mean drag of a circular cylinder at the Reynolds numbers between 40 and 3900. The numerical results showed that the spanwise-periodic blowing/suction attenuated or annihilated the Kármán vortex shedding and thus significantly reduced the mean drag and the drag and lift fluctuations. Dobre et al. (2006) significantly attenuated the spanwise vortex shedding around a square cylinder by using spanwise sinusoidal perturbations with wavelengths of $2.4d$ at the trailing edge at $Re(d) = 12\,500$. Park et al. (2006) achieved 33% base pressure recovery at the Reynolds numbers up to $Re(d) = 60\,000$ using spanwise arrays of vertical rectangular tabs at the trailing edge. Deshpande and Sharma (2012) used segmented trailing edges to achieve up to 34% base pressure recovery and nearly completed the suppression of vortex shedding at $Re(d) = 8750$. More related studies can be found in a review by Choi et al. (2008).

To the authors' knowledge, very few studies have focused on the 3D spanwise-varying control effects on suppressing the VIV of bridges. El-Gammal et al. (2007) successfully suppressed the VIV of a girder by using spanwise periodic perturbation method (SPPM). In their study, the leading and the trailing edges of the deck were modified as a wave in the spanwise direction. However, the shape of the girder was changed dramatically, which could cause esthetic problems and affect other performance aspects. In this paper, the PVGs based on the 3D spanwise-varying control are proposed to attenuate the spanwise vortex shedding and mitigate the VIV of a bridge deck. The PVGs have been widely applied in airfoils to delay the boundary layer separation (Lin, 2002). The streamwise counter-rotating vortex structures (see Fig. 1, Von Stillfried et al., 2010) induced by vortex generators force the higher momentum fluid from the free stream towards the wall and increase the near-wall momentum. Consequently, the stabilised flow leads to the delay or prevention of separation of the boundary layers. As stated above, the streamwise vortices are able to cause the spanwise dislocation of the Karman vortices or even destroy the spanwise vortices. Therefore, the PVGs can be used to generate the streamwise counter-rotating vortex pairs directly and to activate the corresponding mode of secondary instability to suppress the spanwise vortex shedding and mitigate the VIV. Although the PVGs are widely used in boundary layer separation control, their application in controlling the VIV of bluff bodies is missing. Therefore, the main objectives of this study are to propose and design the spanwise-varying PVGs control method and then to investigate its control

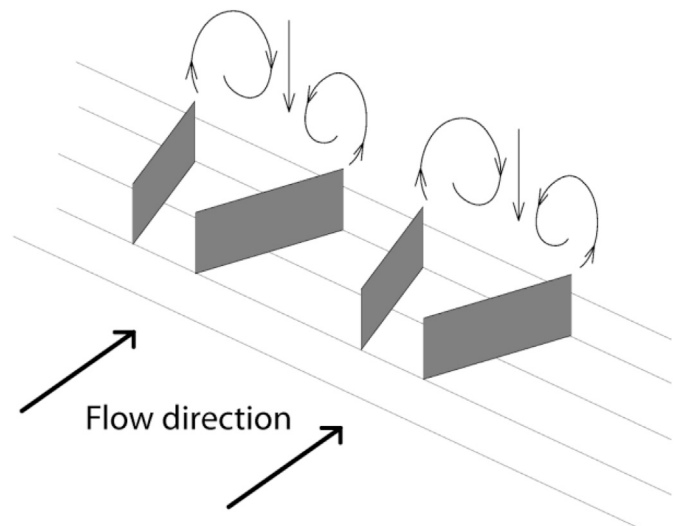


Fig. 1. Generation of streamwise vortex pairs by the PVG (Von Stillfried et al., 2010).

effects on suppressing the VIV of a bridge deck by wind tunnel tests.

2. Experiment setup

The present experiments were conducted in the wind tunnel at the Harbin Institute of Technology in China. The working section size of the wind tunnel was $4\text{ m} \times 3\text{ m} \times 25\text{ m}$ and its wind velocity range was from 2 m/s to 45 m/s . The maximum free stream turbulence intensity was 0.46% and the maximum free-stream non-uniformity was 1% . In this study, the oncoming flow in the tested cases was uniform and smooth. The wind induced motions of the deck model were measured by two laser reflexion sensors shown in Fig. 2, set in pairs on each side of the test section and aimed at reflexive targets set on the lever arms. The signals from the reflexion gauges were sampled at 1000 Hz . The hot wire anemometry (Dantec 55P11) was used to measure the streamwise velocity in the wake and the sample frequency was set to 1000 Hz . The test model was made of acrylic plastic with thickness of 5 mm and was hung on a free vibration device shown in Fig. 2. In order to eliminate the effects of the supporting devices and the end effects of the sectional model, two end plates were installed at the end of the bridge sectional model. A very fine scan of the wind speed range of interest was performed during the test. For every wind speed, data length of 2 min was sampled after the data became steady.

The typical bridge model used in this study is that of the Great Belt East Bridge. The bridge deck was subjected to a severe VIV at a low wind

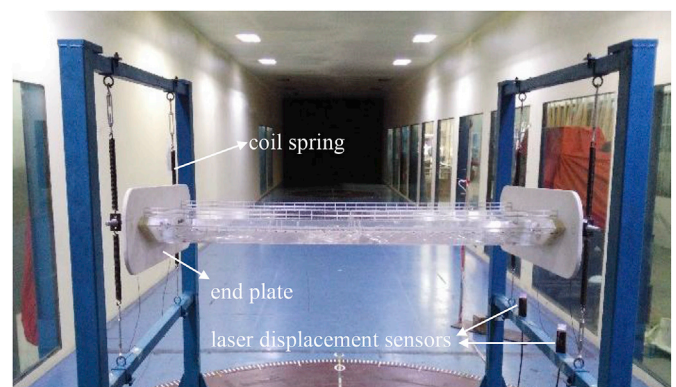


Fig. 2. The suspension system.

speed of 4–12 m/s (Larsen et al., 2000). The size of the model with the scale of 1:40 is shown in Fig. 3 (a), where the width $B = 0.775$ m, the height $H = 0.10$ m and the spanwise length $L = 1.6$ m. The wind tunnel blockage ratio was about 3.3% and its influence was considered negligible. The cases with and without the railing were considered in the wind tunnel tests. The shape and the size of the railing were similar to those used by Larsen and Wall (2012), as shown in Fig. 3 (b). The mass and the mass moment of inertia of the deck sections, including the contribution from the suspension system, were $m = 15.281$ kg/m and $I = 6.802$ kg m²/m, yielding dimensionless values $m/\rho B^2 = 20.75$ and $I/\rho B^4 = 15.38$, where $\rho = 1.226$ kg/m³ is the air density and B is the deck width. Eight identical coil springs were attached symmetrically on the lever arms to provide the bending and torsional stiffness as shown in Fig. 1. The natural vertical and torsional frequencies of the test model in still air were $f_v = 5.249$ Hz and $f_a = 9.613$ Hz respectively, and the damping ratio was $\xi_v = 0.33\%$ and $\xi_a = 0.38\%$.

To achieve better control effects, the PVGs were set close to the separation points (the key region related to spanwise vortex) in the undersurface of the box girder. Because of the uncertainty of the wind direction in natural environment, the PVGs were arranged on the two sides of the test model symmetrically in the test, as shown in Fig. 4. As stated above, the spanwise distance (λ) is the key parameter in controlling the VIV since different spanwise distances correspond to different instability wake modes. For bluff body, Dobre et al. (2006) summarised that the ratio of the spanwise length λ to the streamwise wavelength λ_x was close to a quasi-universal value of 0.63–0.84 for Mode-A. It is noted that the streamwise wavelength $\lambda_x = H/S_t$, where $S_t = fH/U$ (where f is the shedding frequency and U is the free stream velocity). Therefore, the PVGs were set at the spanwise distances 1–5H ($\lambda/\lambda_x = 0.25$ –1.25, here $St = 0.25$), which are the covered quasi-universal values in this study. The arrangement and geometry of the PVGs are presented in Fig. 4, where $h/H = 0.2$, $l_a/H = 0.3$, $l_b/H = 0.7$, $l_c/H = 0.5$.

3. Results and discussion

The results are presented as the plots of vertical or torsional response as function of the non-dimensional wind speed $U/f_v B$ or $U/f_a B$, where U is the wind tunnel speed measured by a hot film anemometer set in the test section, f_v and f_a are the vertical bending frequency (5.25 Hz) and torsional frequency (9.61 Hz) respectively, B (775 mm) is the deck width. The vertical vibration displacement, y , is given as the root mean square (RMS) vertical response normalised by the section height H . The torsional vibration displacement, α , is given as the root mean square (RMS) torsional response. Both of the cases with and without the railing were tested. In addition, all the S_t values were obtained by using the Fast Fourier Transformation (FFT) of the wake velocity (measured by hot wire anemometry) when the bridge deck was static.

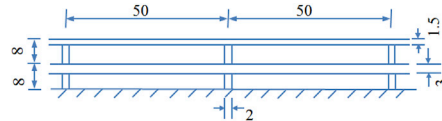
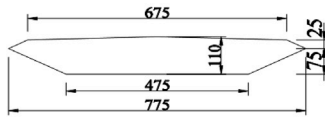


Fig. 3. Size of (a) the test model and (b) the railing (unit: mm).

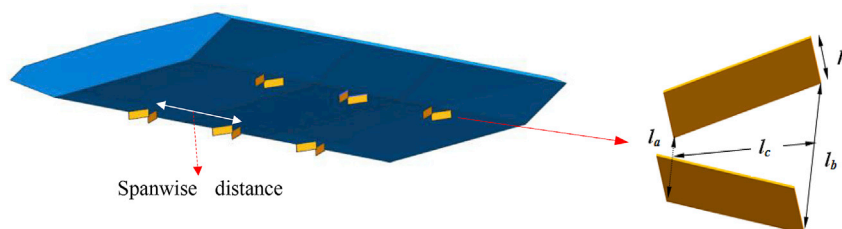


Fig. 4. Arrangement of the PVGs and its parameters.

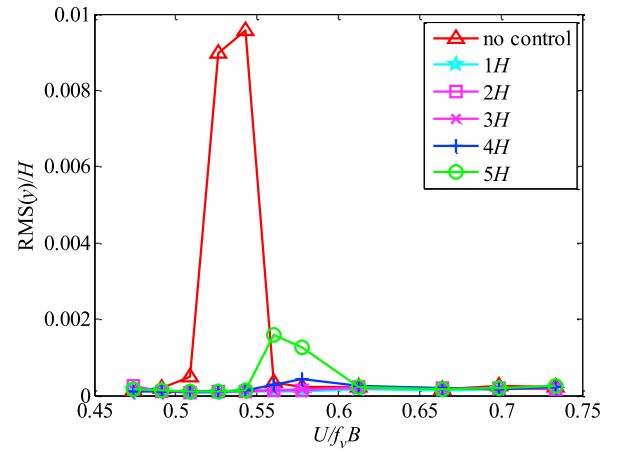


Fig. 5. RMS of the vertical response against the reduced wind velocity under different spanwise distances ($h/H = 0.2$).

3.1. Without railing

(1) Control effects of the spanwise distance

The control effects of the PVG on reducing the vortex induced vertical and torsional response with different spanwise distances are shown in Fig. 5 and Fig. 6 respectively, where 1–5H represents the spanwise distances of 1–5H.

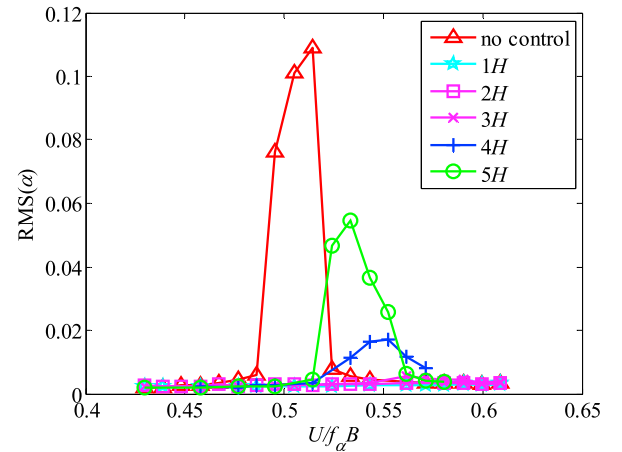


Fig. 6. RMS of the torsional response against the reduced wind velocity under different spanwise distances ($h/H = 0.2$).

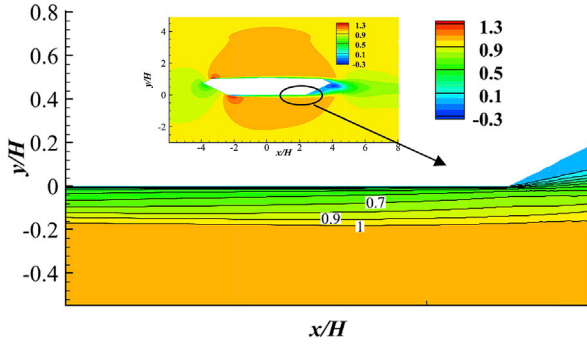


Fig. 7. Mean streamwise velocity contours (Note: the velocity value is normalised by the oncoming flow velocity).

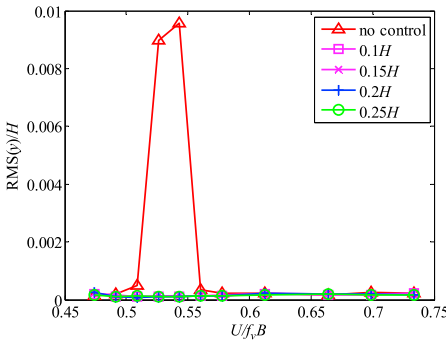
The vertical response peak at $U/f_v B \approx 0.54$ is within the bending lock-in range without railing (Larsen and Wall, 2012). As shown in Fig. 5, the cases under different spanwise distances have different control effects, implying that the control effects are sensitive to the spanwise distance. The vortex induced response is completely suppressed under the spanwise distance of $1-4H$. The ratio of the spanwise distance of $3H$ to the streamwise wavelength of trailing vortex $\lambda/\lambda_x = 0.744$ ($\lambda_x = H/S_L$), which is located within the range $0.63-0.84$ associated with the Mode-A instability (Dobre et al., 2006). As discussed in the introduction, the PVG can directly generate streamwise vortex pairs which activate the Mode-A instability and suppress the trailing edge spanwise vortices. Therefore, it can be the main reason that the VIV is suppressed by the PVG. It is noted that under spanwise distances of $1H$, $2H$ and $4H$, the PVGs also show great control effects, and obviously, the ratios of $\lambda/\lambda_x = 0.248$, 0.496 and 0.992 are outside the range $0.63-0.84$. This indicates that the wake instability can be activated at a wider range of spanwise distance.

The torsional response peak at $U/f_a B \approx 0.52$ ($U/f_v B \approx 0.95$) is within the torsional lock-in range without railing (Larsen and Wall, 2012). As shown in Fig. 6, the control effects of PVG on suppressing torsional VIV are sensitive to the spanwise distance. The vortex induced torsional response is completely suppressed under the spanwise distance of $1-3H$. Similar to the control effect on mitigating vertical response, the PVGs also show great control effects for the spanwise distances of $1-4H$. Obviously, the ratios of $\lambda/\lambda_x = 0.252-1.01$ are also out of the range $0.63-0.84$. This also indicates that the secondary wake instability can be activated at a wider range of spanwise intervals. In addition, the onset wind velocity as well as the peak wind velocity of the VIV is raised, which indicates that the PVG delays the trailing vortex shedding.

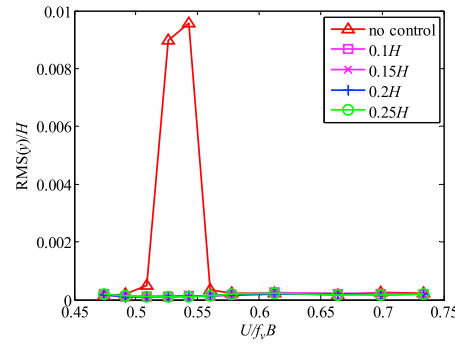
The Reynolds number range in this experiment was $1.04 \times 10^5 - 2.49 \times 10^5$ (based on the width B). The almost constant Strouhal number $S_L \approx 0.25$ measured by hot wire anemometry at this Reynolds number range agrees well with the result of $S_L = 0.254$ (based on the height H , naked deck) at $Re = 2.11 \times 10^5$ by Taylor et al. (2009). The corresponding reduced wind velocity $U_r = (H/B) \cdot (1/S_L) = 0.52$, which is consistent with that of the peak vertical and torsional response. It indicates that the Strouhal number S_L is insensitive to the Reynolds number at this range, and the VIV is induced by wake vortex shedding. The PVG is effective on suppressing VIV in this Reynolds number range. Nevertheless, the Reynolds number of the full scale bridge deck is much higher, and the control effects of the PVG in the real world need to be further studied.

(2) Control effects of the PVG height

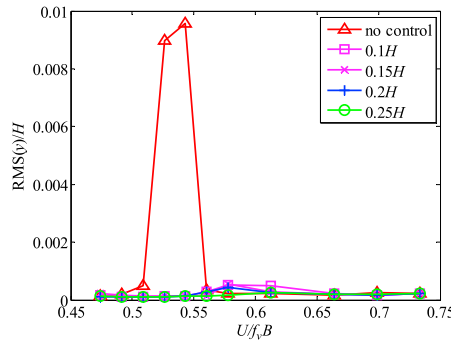
Four cases with different heights of the PVGs ($h = 0.1H$, $0.15H$, $0.2H$ and $0.25H$) were designed to study their control effects on the VIV according to the boundary layer thickness. The boundary layer thickness of the bridge model was obtained by the previous numerical study (Zhang



(1) $\lambda/H=2$



(2) $\lambda/H=3$



(3) $\lambda/H=4$

Fig. 8. RMS of the vertical response vs. the reduced wind velocity with different heights of PVG at spanwise distance of (a) $\lambda/H = 2$, (b) $\lambda/H = 3$ and (c) $\lambda/H = 4$.

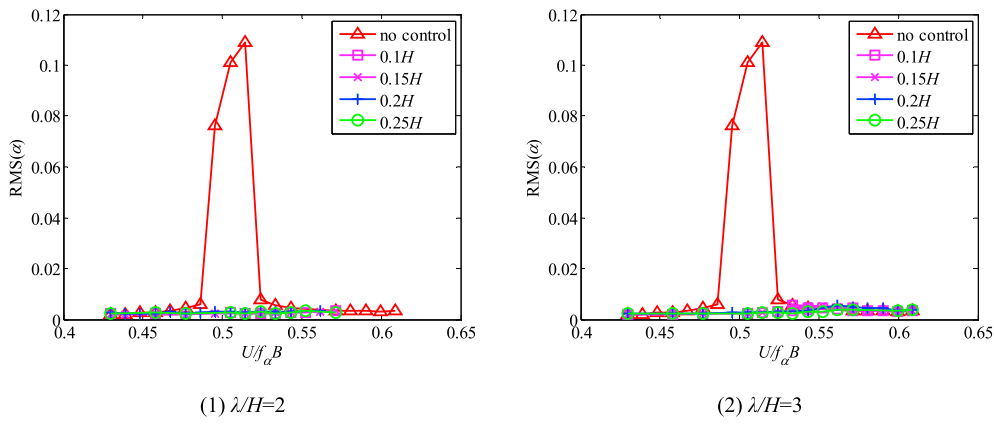


Fig. 9. RMS of the torsional response vs. the reduced wind velocity for different heights of the PVGs at the spanwise distance of (a) $\lambda/H = 2$, (b) $\lambda/H = 3$ and (c) $\lambda/H = 4$.

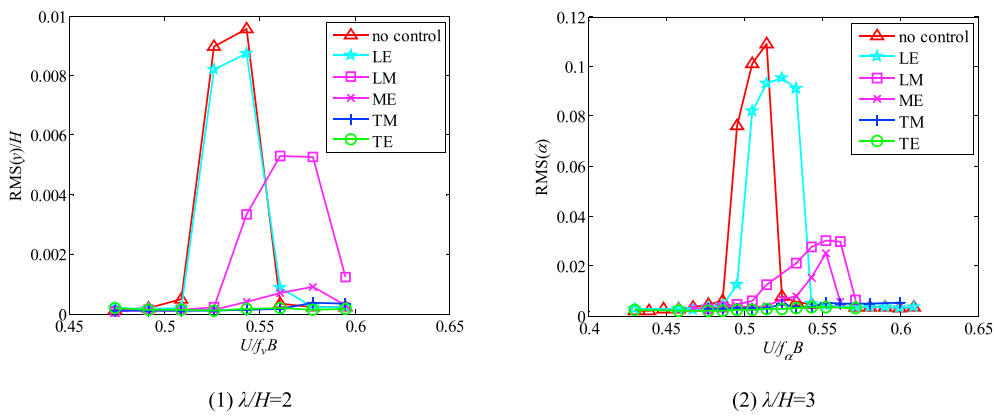
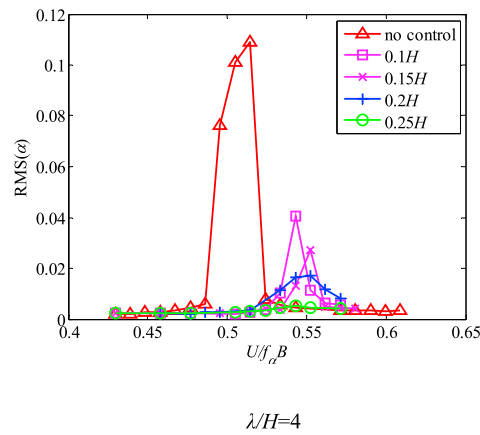


Fig. 10. RMS of (a) the vertical response and (b) the torsional response vs. the non-dimensional wind velocity under different streamwise location ($\lambda/H = 3$, $h/H = 0.25$). (c) sketch of the streamwise locations of PVGs.



(3) $\lambda/H=4$

et al., 2016). Here, the boundary layer thickness is the distance from the wall to a point where the flow velocity reaches 99% of the free stream velocity. The time-averaged streamwise velocity contours are shown in Fig. 7, where the velocity value is normalised by free stream velocity. In Fig. 7, the boundary layer thickness is about $0.17H$; hence, the cases with the PVG height $h = 0.1H$ and $h = 0.15H$ are immersed in the boundary layer, while the cases $h = 0.2H$ and $h = 0.25H$ are bulged out of the boundary layer.

The RMS of the vertical response versus the reduced wind velocity with different heights of PVG is shown in Fig. 8. It is noted that in both cases the PVG within and extended out of the boundary layer completely suppress the VIV. Unlike the conventional control method using the guide vanes (GV), the control effects of the PVG are not influenced by the boundary layer thickness (at least in the range of $0.1\text{--}0.25H$), while the GVs have no control effects on mitigating the VIV at low Reynolds number because no sufficient flow rate passes it with this high boundary layer (Larsen et al., 2008).

The RMS of the torsional response versus the reduced wind velocity with different heights of the PVG is shown in Fig. 9. The PVG cases at the spanwise distances of $\lambda/H = 2$ and $\lambda/H = 3$ completely suppress the VIV and also show that the control effects are uncorrelated with the PVG height. For the PVG cases at the spanwise distance of $\lambda/H = 4$, the RMS of torsional responses are dramatically reduced compared to the base case and the torsional responses decrease with the increase of the PVGs height. Moreover, the onset wind velocity and the peak wind velocity of the VIV are clearly raised.

(3) Control effects of the installed positions

Five positions in the streamwise direction are labelled on the undersurface of the deck, as shown in Fig. 10 (c). In Fig. 10 (c), the case LE indicates that the PVGs are only set in the leading edge on the

undersurface; TE represents that the PVGs are only set in the trailing edge on the undersurface; ME represents that the PVGs are set in the middle of the leading edge and the trailing edge of the undersurface; LM represents that the PVGs are arranged in the middle of the LE and the ME; and TM represents that the PVGs are arranged in the middle of the TE and the ME.

The control effects are highly sensitive to the streamwise position of the PVG. As the PVGs were arranged closer to the trailing edge (TE), the RMS of the vertical and torsional response are smaller. When the PVG is arranged in the leading edge (LE), the vibration amplitude is not mitigated and even the range of the reduced wind velocity is slightly wider than the baseline case. While the PVG is arranged in the trailing edge (TE), the vibration amplitude is dramatically mitigated, which indicates that the trailing edge is the key region in controlling the VIV and thus, the trailing edge spanwise vortices are the main reason in inducing the VIV. Furthermore, the PVGs arranged in TE, which creates streamwise vortex pairs, can destroy or prevent the development of spanwise trailing edge vortices. Here, it can be speculated that as the PVGs are arranged in LE, the generated streamwise vortex pairs are damped out in the shear layer while they arrive at the wake. Therefore, the trailing edge spanwise vortices are not influenced by the streamwise vortex pairs and thus the vortex induced responses are not mitigated. In addition, as the PVGs were arranged closer to the trailing edge (TE), the onset wind velocity as well as the peak wind velocity of VIV are raised, which indicates that the PVG delays the trailing vortex shedding.

(4) Control effects influenced by the wind attack angles

The control effects of the PVG on mitigating the VIV under the wind attack angle of $\pm 3^\circ$ are shown in Fig. 11, which includes the plots of the RMS of vertical and torsional response as function of the reduced wind speed at different spanwise distance. Compared to the wind attack angle of 0° , the peak RMS values of the vertical and torsional responses are

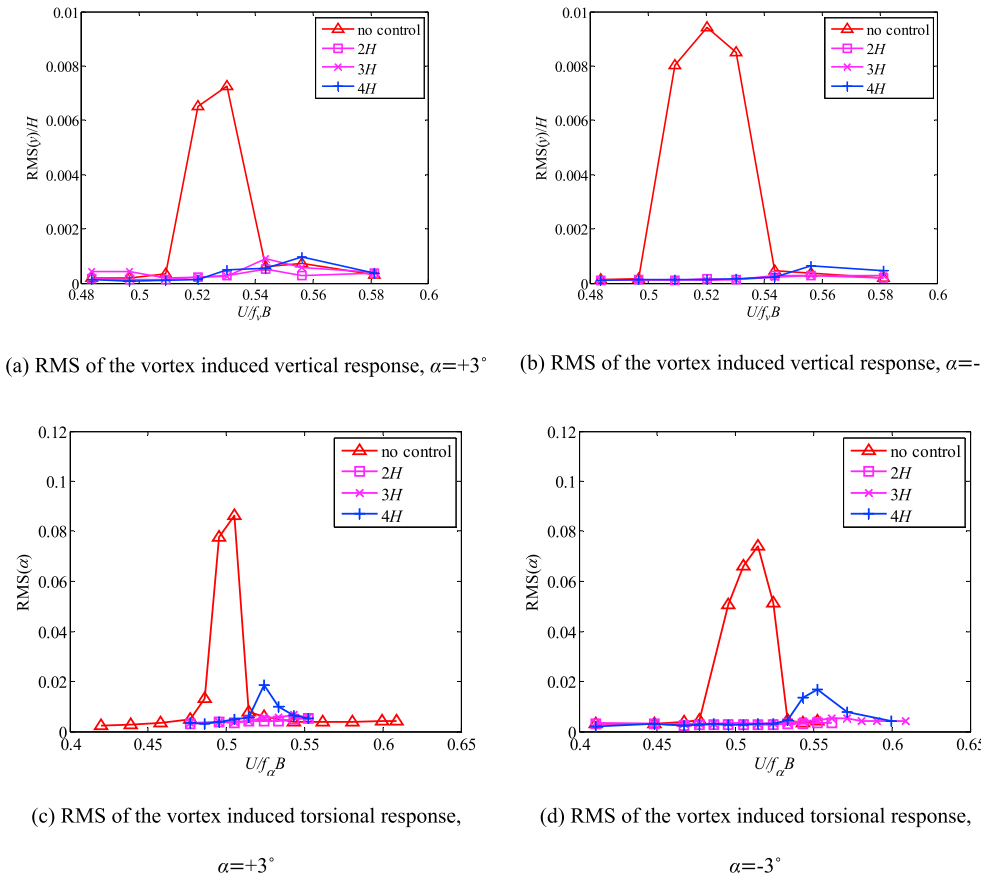


Fig. 11. Control effects of the PVGs with different spanwise distances under the wind attack angle of $\pm 3^\circ$ ($h/H = 0.15$).

slightly decreased at the wind attack angle of $\pm 3^\circ$. The vortex induced vertical response is completely suppressed by using the PVGs with spanwise distance of $2\text{--}4H$, which show similar control effects to that under the wind attack angle of 0° . The vortex induced torsional response is also completely suppressed by using the PVGs with spanwise distance of $2H$ and $3H$. For the PVG case with spanwise distance of $4H$, the torsional response is dramatically reduced and the onset wind velocity and the peak wind velocity of VIV are also raised, thus showing a similar control effect to that under the wind attack angle of 0° . Generally, the results show that both vortex induced vertical and torsional responses can be completely suppressed by using the PVG under the wind attack angle of $\pm 3^\circ$.

(5) Wake analysis

The wake decay analysis was based on the investigation of the downstream evolution of the second-order turbulence statistics and on the PSD. Fig. 12 shows the downstream evolution of the RMS value of the streamwise velocity u'/U_0 for the four wake locations of $x/H = 7, 12, 17$ and 21 , where u' is the RMS of fluctuating streamwise velocity and U_0 is the mean local velocity. The base case is the case without control, and $z = 0$ and $z = 1.5H$ are the two spanwise measurement locations. The diagram of the coordinate system is shown in Fig. 12(c), while Fig. 13 shows the PSD of the vortex induced responses and the streamwise velocity in the wake. For the vertical response, PSD is normalised by the vertical frequency and the bridge deck height; for the torsional response, PSD is normalised by the vertical frequency; and for the wake velocity, PSD is normalised by the vertical frequency and the free stream velocity.

Fig. 12 shows that for the base case, the RMS of the fluctuating

streamwise velocity decreases as the flow progresses downstream, which indicates the wake decay. For the PVG cases, not only the RMS of fluctuating streamwise velocity is considerably lower, but the decreased slopes are also greater when compared to those of the base case. The results implies that the arranged PVGs may lead to the suppression of the motion or flow instability induced trailing edge vortices (TEV) in wake and may also cause the wake decay to accelerate. It is noted that there is no difference between the two spanwise locations (i.e., $z = 0$ and $z = 1.5H$), implying that the wake is statistically homogeneous in this direction. Therefore, it is indicated that the effects of PVG are beneficial in the reduction of the streamwise turbulence intensity at all spanwise locations.

In Fig. 13(a), the vortex-induced vibration frequency is $f = f_v = 5.25$ (i.e., $f_v H/U = 0.238$) at $U/f_v B = 0.54$ for the base case, while the VIV is completely suppressed by the PVGs shown in Fig. 13(b). The domain streamwise velocity frequency in the wake, which has the same value as the vibration frequency, shows some deviation from the vortex shedding frequency (i.e., $f_v H/U = 0.248$) of the static bridge deck. Meanwhile, the double frequency of streamwise velocity is also observed in the wake. It implies that the vortex shedding frequency is locked by the natural frequency of the bridge deck. The PSD peak of the streamwise velocity disappears under the control of the PVG as shown in Fig. 13 (d), which confirms that the motion or flow instability induced TEV are destroyed by the PVG. Similar results can be obtained for the vortex induced torsional response ($U/f_v B = 0.50$), as shown in Fig. 13(e) ~ (h). Interestingly, in Fig. 12(g) the two peak values of the streamwise velocity PSD are detected in the base case and the peak value of the double frequency is higher than that of the fundamental frequency, but both peaks vanish following the use of the PVGs.

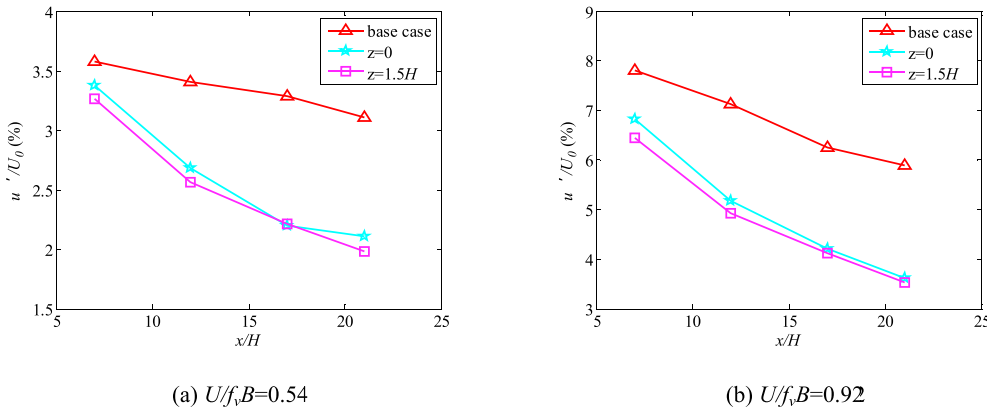
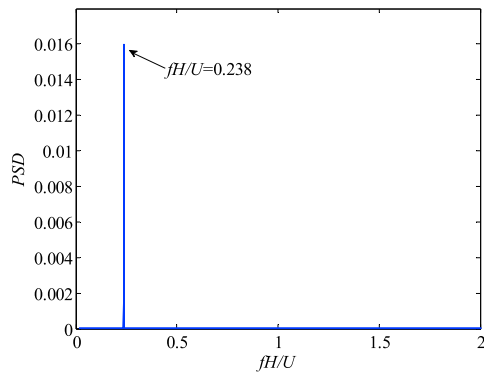
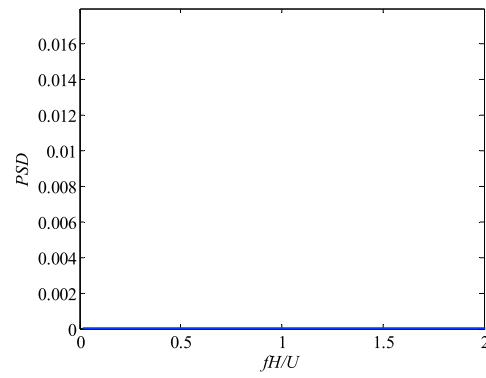
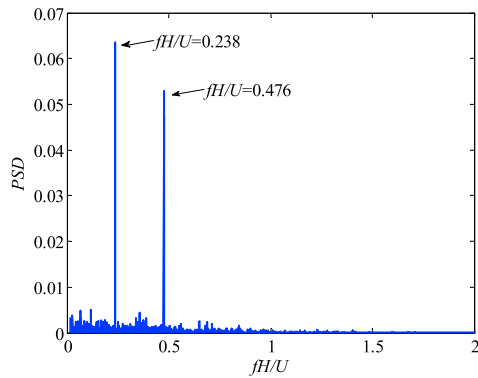
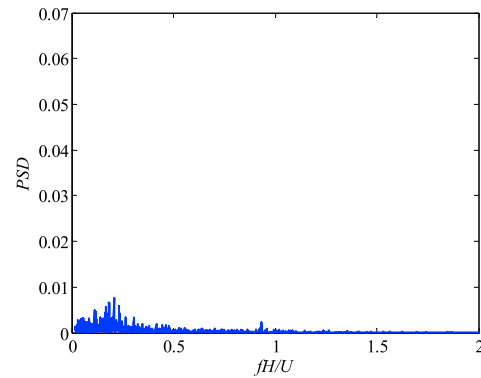
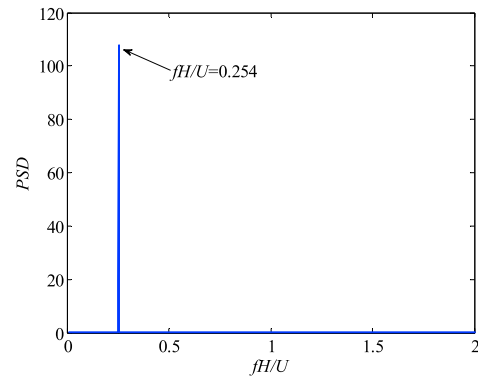
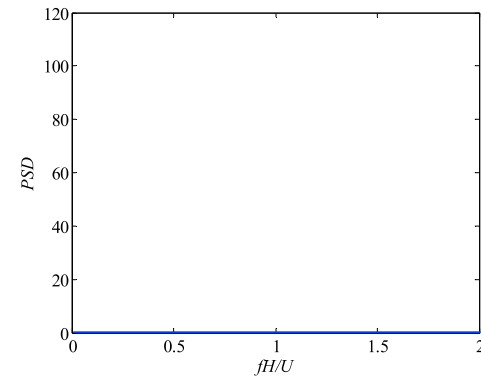
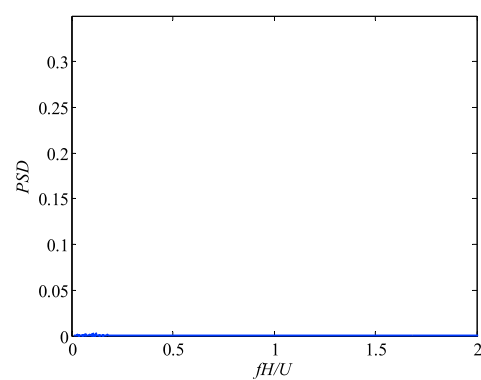
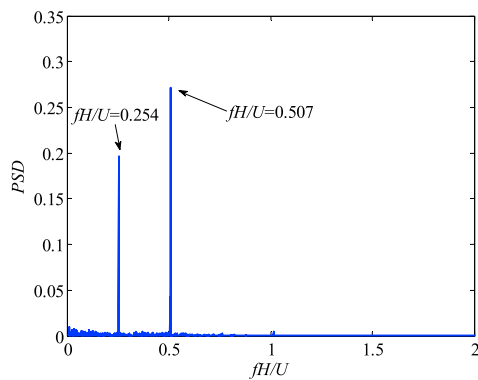


Fig. 12. Downstream evolution of the RMS value of the streamwise velocity ($\lambda/H = 3$).

(a) Base case, vertical response, $U/f_v B = 0.54$ (b) $\lambda/H = 3$, $h/H = 0.25$, vertical response, $U/f_v B = 0.54$ (c) Base case, streamwise velocity, $U/f_v B = 0.54$ (d) $\lambda/H = 3$, $h/H = 0.25$, streamwise velocity, $U/f_v B = 0.54$ (e) Base case, torsional response, $U/f_\alpha B = 0.50$ (f) $\lambda/H = 3$, $h/H = 0.25$, torsional response, $U/f_\alpha B = 0.50$ Fig. 13. PSD of the vortex induced responses and the streamwise velocity in wake ($x/H = 7$, $z/H = 0$).

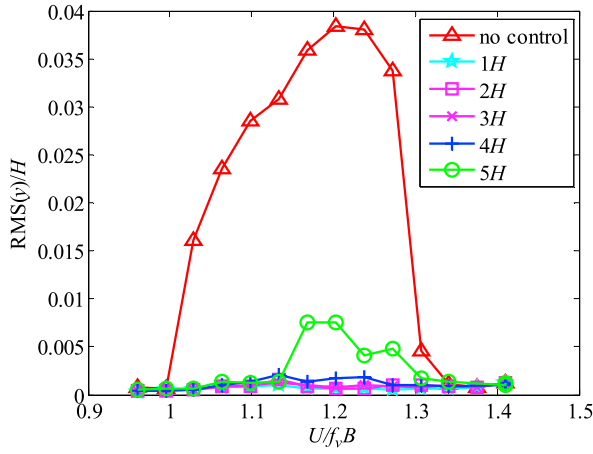


Fig. 14. RMS of the vertical response against the reduced wind velocity with different spanwise distances (with railing, $h/H = 0.2$).

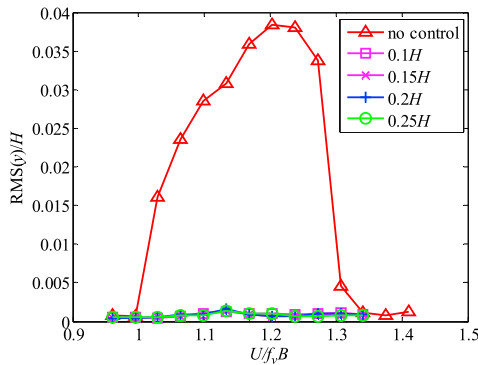
3.2. With railing

(1) Control effects of the spanwise distance

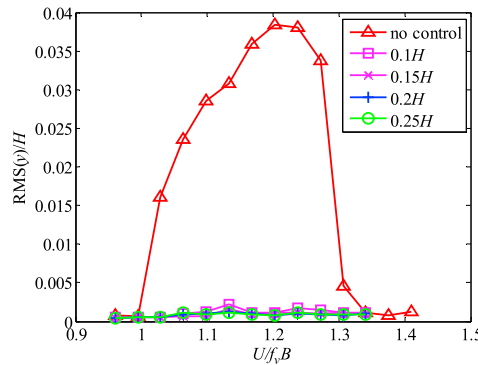
For the cases with the railing, the control effects on the VIV of the PVG with different spanwise distances are shown in Fig. 14, where 1–5H respectively represent the spanwise distances of 1–5H. For the base case (i.e., no control) with the railing, only the vortex induced vertical vibration in the range of $U/f_v B = 1$ –1.35 is detected in the range of $U/f_v B = 0.9$ –2.1 (not present). The amplitude of the VIV with the railing is significantly higher than that without the railing. The vertical response peak at $U/f_v B \approx 1.2$ agrees well with the previous investigations (Larsen and Wall, 2012). As shown in Fig. 14, the vortex induced torsion response is completely suppressed under the PVGs spanwise distance of 1–4H, which shows excellent control effects on mitigating vertical response similar to that of the naked deck. In addition, the onset wind velocity of the VIV is raised, which also indicates that the PVG delays the vortex shedding.

(2) Control effects of the PVG height

The RMS of the vertical response (with railing) versus the reduced wind velocity with different heights of the PVGs is shown in Fig. 15. The PVG cases at the spanwise distance of $\lambda/H = 2$ and $\lambda/H = 3$ completely suppress the VIV and show that the control effects are independent of the PVG height, at least in the range of the present PVG heights.



(a) $\lambda/H = 2$



(b) $\lambda/H = 3$

(3) Control effects of the installed positions

Three different positions in the streamwise direction are labelled in the undersurface, as shown in Fig. 16. The signs LE, ME and TE have the same meaning as in Fig. 10 (c). The control effects are sensitive to the streamwise position of the PVG, which shows same results as the naked bridge deck in Fig. 10. As the PVGs are arranged closer to TE, the RMS of the vertical is highly reduced. As the PVG is arranged in the trailing edge (TE), the VIV is completely suppressed, implying the trailing edge is also the key region in controlling the VIV, and thus, the spanwise vortices near the trailing edge are also the main reason inducing the VIV. Further, the PVGs arranged in the TE, which creates streamwise vortex pairs, can destroy or prevent the development of spanwise trailing edge vortices. In addition, as the PVGs are arranged closer to the TE, the onset wind velocity and the peak wind velocity of the VIV are raised, indicating that the PVG delays the trailing vortex shedding.

(4) Control effects influenced by the wind attack angles

The control effects of PVG on mitigating the VIV under the wind attack angle of $\pm 3^\circ$ (with the railing) are shown in Fig. 17, which presents the plot of the RMS of the vertical response as a function of the reduced wind velocity at different spanwise distances. Compared to the wind attack angle of 0° , the peak RMS values of the vertical responses increase and decrease slightly under the wind attack angle of $+3^\circ$ and -3° respectively. The vortex induced vertical response is also completely suppressed by using the PVGs with the spanwise distance of 2H and 3H,

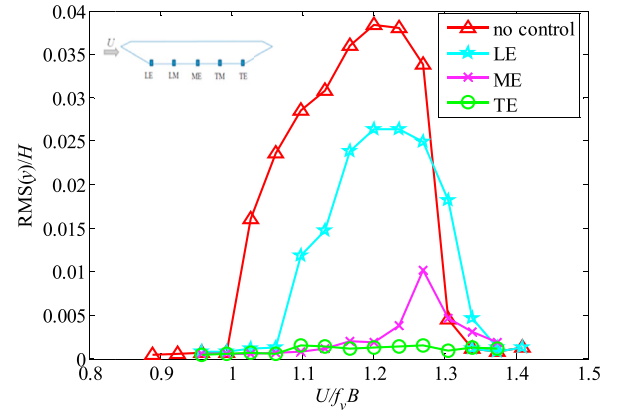


Fig. 16. RMS of vertical the response vs. the non-dimensional wind velocity under different streamwise location ($\lambda/H = 3$, $h/H = 0.25$).

Fig. 15. RMS of the vertical response vs. the reduced wind velocity with different heights of PVG at spanwise distance of (a) $\lambda/H = 2$ and (b) $\lambda/H = 3$ (with railing).

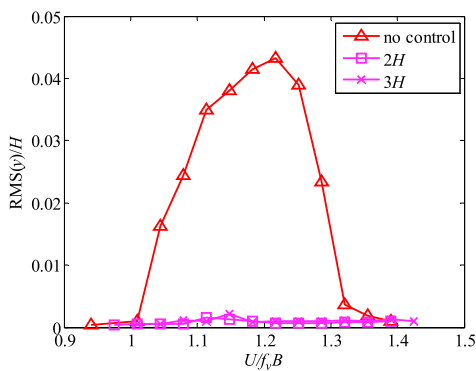
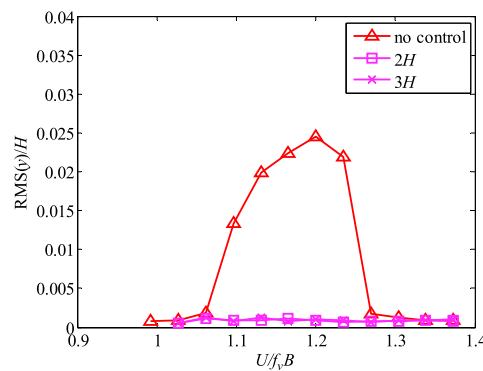
(a) $\alpha=+3^\circ$ (b) $\alpha=-3^\circ$

Fig. 17. Control effects of the PVGs under the wind attack angle of $\pm 3^\circ$ (with railing, $h/H = 0.15$).

which also shows excellent control effects similar to that under the wind attack angle of 0° .

4. Conclusions

Because of the high efficiency of 3D spanwise-varying method on suppressing vortex shedding behind the bluff body, the PVG, which can generate the streamwise vortices, was adopted to mitigate the VIV of a bridge. The wind tunnel tests were conducted to prove the control effects of the PVG on the VIV of both the naked bridge deck and the deck with railings. Major conclusions drawn are as follows:

- 1) For the naked bridge deck, the spanwise-distributed PVGs with interval of $1-4H$ and $1-3H$ completely suppress the vortex induced vertical and torsion response. As the PVGs are arranged at the interval of $5H$ and $4-5H$, the VIV can be also dramatically mitigated. The control effects of the PVG are not influenced by the boundary layer height in the range of $0.1-0.25H$. As the PVGs were arranged closer to the TE, the vertical and torsional responses are smaller and the onset wind velocity and the peak wind velocity of the VIV are also raised, implying that the trailing edge is the key region in controlling the VIV, and thus, the trailing edge spanwise vortices may be also the main reason inducing the VIV. In addition, both vortex induced vertical and torsional responses can be completely suppressed by using the PVG under the wind attack angle of $\pm 3^\circ$.
- 2) A violent VIV is also introduced as the bridge is installed with the railing and the amplitude is considerably higher than that of the naked bridge. The PVGs at the intervals of $1-4H$ also completely suppress the VIV and the control effects are not influenced by the heights of the PVG in the range of $0.1-0.25H$. As the PVGs are arranged at the interval of $5H$, the VIV is also dramatically mitigated and the onset wind velocity of VIV is raised, which also indicates that the PVG delays the vortex shedding. Similar to the case of the naked bridge, as the PVGs are arranged closer to TE, the vertical response is also smaller, implying the important role of the TE in mitigation of the VIV. In addition, under the wind attack angle of $\pm 3^\circ$, the vortex induced vertical response can be completely suppressed by using the PVG.
- 3) The wake decay and the spectrum analysis of the wake velocity in free vibration cases show that as the PVGs are arranged on the bridge, the PSD peak of the wake velocity is suppressed, the fluctuating wind velocity is reduced at all spanwise locations and the wake decay is accelerated. The results show that the PVG can suppress the motion or flow instability induced vortex shedding near the wake and thus result in the vanishing of the VIV.

As mentioned above, the PVGs have been widely applied to the airfoils to delay the flow separation (Lin, 2002). The flow separation is one

of the sources leading to the flutter or galloping of a bridge. Therefore, the PVG may have advantageous effects on improving the flutter or the galloping instability. Although the PVG is expected to suppress regular vortex shedding and thus reduce the pressure-induced drag, the PVG elements may increase substantially the mean drag force. The studies on the aerodynamic properties and the flutter properties of the deck with the PVG are our future work.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.jweia.2018.01.046>.

References

- Battista, R.C., Pfeil, M.S., 2000. Reduction of vortex-induced oscillations of Rio-Niterói bridge by dynamic control devices. *J. Wind Eng. Ind. Aerod.* 84 (3), 273–288. [https://doi.org/10.1016/S0167-6105\(99\)00108-7](https://doi.org/10.1016/S0167-6105(99)00108-7).
- Choi, H., Jeon, W.P., Kim, J., 2008. Control of flow over a bluff body. *Annu. Rev. Fluid Mech.* 40, 113–139. <https://doi.org/10.1146/annurev.fluid.39.050905.110149>.
- Darekar, R.M., Sherwin, S.J., 2001. Flow past a square-section cylinder with a wavy stagnation face. *J. Fluid Mech.* 426, 263–295. <https://doi.org/10.1017/S0022112000002299>.
- Deshpande, P.J., Sharma, S.D., 2012. Spanwise vortex dislocation in the wake of segmented blunt trailing edge. *J. Fluid Struct.* 34, 202–217. <https://doi.org/10.1016/j.jfluidstructs.2012.04.004>.
- Dobre, A., Hangan, H., Vickery, B.J., 2006. Wake control based on spanwise sinusoidal perturbations. *AIAA J.* 44 (3), 485–492. <https://doi.org/10.2514/1.15155>.
- Doddipatla, L.S., Hangan, H., Durgesh, V., Naughton, J.W., 2008. Wake energy redistribution due to trailing edge spanwise perturbation. In: *Proceedings of the BBAA VI International Colloquium on Bluff Bodies Aerodynamics and Applications*. July. Milano, Italy, pp. 20–24.
- El-Gammal, M., Hangan, H., King, P., 2007. Control of vortex shedding-induced effects in a sectional bridge model by spanwise perturbation method. *J. Wind Eng. Ind. Aerod.* 95 (8), 663–678. <https://doi.org/10.1016/j.jweia.2007.01.006>.
- Frandsen, J.B., 2001. Simultaneous pressures and accelerations measured full-scale on the Great Belt East suspension bridge. *J. Wind Eng. Ind. Aerod.* 89 (1), 95–129. [https://doi.org/10.1016/S0167-6105\(00\)00059-3](https://doi.org/10.1016/S0167-6105(00)00059-3).
- Fujino, Y., Siringoringo, D., 2013. Vibration mechanisms and controls of long-span bridges: a review. *Struct. Eng. Int.* 23 (3), 248–268. <https://doi.org/10.2749/101686613X13439149156886>.
- Fujino, Y., Yoshitaka, Y., 2002. Wind-induced vibration and control of trans-tokyo bay crossing bridge. *J. Struct. Eng.* 128 (8), 1012–1025. [https://doi.org/10.1061/\(ASCE\)0733-9445\(2002\)128:8\(1012\)](https://doi.org/10.1061/(ASCE)0733-9445(2002)128:8(1012)).
- Hwang, Y., Kim, J., Choi, H., 2013. Stabilization of absolute instability in spanwise wavy two-dimensional wakes. *J. Fluid Mech.* 727, 346–378. <https://doi.org/10.1017/jfm.2013.270>.

- Kim, J., Choi, H., 2005. Distributed forcing of flow over a circular cylinder. *Phys. Fluids* 17 (3), 033103. <https://doi.org/10.1063/1.1850151>.
- Larsen, A., Wall, A., 2012. Shaping of bridge box girders to avoid vortex shedding response. *J. Wind Eng. Ind. Aerod.* 104, 159–165. <https://doi.org/10.1016/j.jweia.2012.04.018>.
- Larsen, A., Esdahl, S., Andersen, J.E., Vejrum, T., 2000. Storebaelt suspension bridge–vortex shedding excitation and mitigation by guide vanes. *J. Wind Eng. Ind. Aerod.* 88 (2), 283–296. [https://doi.org/10.1016/S0167-6105\(00\)00054-4](https://doi.org/10.1016/S0167-6105(00)00054-4).
- Larsen, A., Savage, M., Lafrenière, A., Hui, M.C.H., Larsen, S.V., 2008. Investigation of vortex response of a twin box bridge section at high and low Reynolds numbers. *J. Wind Eng. Ind. Aerod.* 96 (6), 934–944. <https://doi.org/10.1016/j.jweia.2007.06.020>.
- Lin, J.C., 2002. Review of research on low-profile vortex generators to control boundary-layer separation. *Prog. Aerosp.* 38 (4), 389–420.
- Park, H., et al., 2006. Drag reduction in flow over a two-dimensional bluff body with a blunt trailing edge using a new passive device. *J. Fluid Mech.* 563, 389–414. <https://doi.org/10.1017/S0022112006001364>.
- Taylor, Z., Gurka, R., Kopp, G.A., 2009. Geometric effects on shedding frequency for bridge sections. In: *Proceedings of the 11th Americas Conference on Wind Engineering*. San Juan, Puerto Rico.
- Von Stillfried, F., Wallin, S., Johansson, A.V., 2010. An improved passive vortex generator model for flow separation control. *AIAA J.* <https://doi.org/10.2514/6.2010-509>.
- Williamson, C.H.K., 1996a. Vortex dynamics in the cylinder wake. *Annu. Rev. Fluid Mech.* 28 (1), 477–539. <https://doi.org/10.1146/annurev.fl.28.010196.002401>.
- Williamson, C.H.K., 1996b. Three-dimensional wake transition. *J. Fluid Mech.* 328, 345–407. <https://doi.org/10.1017/S0022112096008750>.
- Wu, J., Sheridan, J., Welsh, M.C., Hourigan, K., 1996. Three-dimensional vortex structures in a cylinder wake. *J. Fluid Mech.* 312, 201–222. <https://doi.org/10.1017/S0022112096001978>.
- Zhang, H., Xin, D., Ou, J., 2016. Wake control of vortex shedding based on spanwise suction of a bridge section model using delayed detached eddy simulation. *J. Wind Eng. Ind. Aerod.* 155, 100–114. <https://doi.org/10.1016/j.jweia.2016.05.004>.